

GEOTECHNICAL ENGINEERING REPORT



Lidl – Norcross
4990 Jimmy Carter Boulevard
Norcross, Gwinnett County, Georgia

PREPARED FOR:

Lidl US, LLC
400 Interstate N. Parkway SE, Suite 320
Atlanta, Georgia 30339

NOVA Project Number: 10103-2021006

March 26, 2021



March 26, 2021

Lidl US, LLC

400 Interstate N. Parkway SE, Suite 320
Atlanta, Georgia 30339

Attention: Mr. Delbert Renfro
Real Estate Manager

Subject: Subsurface Exploration and Geotechnical Engineering Report
LIDL – NORCROSS
4990 JIMMY CARTER BOULEVARD
Norcross, Gwinnett County, Georgia
NOVA Project Number 10103-2021006

Dear Mr. Renfro:

NOVA Engineering and Environmental, LLC (NOVA) has completed the authorized Subsurface Exploration and Geotechnical Engineering Report for the proposed Lidl – Norcross development at 4990 Jimmy Carter Boulevard in Norcross, Georgia. The work was performed in general accordance with NOVA Proposal Number 003-20211025 dated February 11, 2021. This report briefly discusses our understanding of the project at the time of the subsurface exploration, describes the geotechnical consulting services provided by NOVA, and presents our findings, conclusions, and recommendations.

We appreciate your selection of NOVA and the opportunity to be of service on this project. If you have any questions, or if we may be of further assistance, please do not hesitate to contact us.

Sincerely,
NOVA Engineering and Environmental, LLC

Michael W. McKenzie, E.I.T.
Project Engineer



Marc D. Johnston, P.E.
Regional Manager
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Copies Submitted: Addressee (electronic)

TABLE OF CONTENTS

1.0	INTRODUCTION	1
1.1	PROJECT INFORMATION.....	1
1.2	SCOPE OF WORK	2
2.0	SITE DESCRIPTION.....	4
2.1	LOCATION AND LEGAL DESCRIPTION	4
2.2	SUBJECT PROPERTY AND VICINITY GENERAL CHARACTERISTICS.....	4
2.3	CURRENT USE OF THE PROPERTY	4
3.0	FIELD AND LABORATORY PROCEDURES	5
3.1	FIELD EXPLORATION	5
3.2	LABORATORY TESTING.....	6
4.0	SUBSURFACE CONDITIONS	9
4.1	GEOLOGY	9
4.2	SOIL AND ROCK CONDITIONS.....	10
4.3	GROUNDWATER CONDITIONS	11
4.4	PAVEMENT ANALYSIS.....	12
5.0	CONCLUSIONS AND RECOMMENDATIONS	14
5.1	SITE PREPARATION	14
5.2	FILL PLACEMENT	16
5.3	GROUNDWATER CONTROL	17
5.4	SLOPES	17
5.5	FOUNDATIONS.....	18
5.6	SLAB-ON-GRADE.....	20
5.7	BELOW GRADE WALLS	21
5.8	PAVEMENT SECTIONS.....	22
5.9	SEISMIC SOIL SITE CLASS.....	25
6.0	CONSTRUCTION OBSERVATIONS	26
6.1	SUBGRADE	26
6.2	SHALLOW FOUNDATIONS	26

APPENDICES

Appendix A – Figures and Maps
Appendix B – Subsurface Data
Appendix C – Laboratory Test Results
Appendix D – Qualifications of Recommendations

1.0 INTRODUCTION

1.1 PROJECT INFORMATION

Our understanding of the requirements of the project is based on our conversations with Mr. Delbert Renfro of Lidl US, LLC, and review of the provided project documents, as well as our experience with similar projects. The subject property is located at 4990 Jimmy Carter Boulevard in Norcross, Gwinnett County, Georgia.

1.1.1 Site Plans and Documents

We were furnished with the following plans and documents:

- Overall Site Plan, prepared by CHA, dated November 10, 2020.

1.1.2 Proposed Structures

We understand proposed construction currently consists of a grocery store building with associated parking lots, driveways, and other infrastructure. The development will consist of an addition and renovation to an existing warehouse building to convert it to a Lidl grocery store. The building, inclusive of the proposed building addition, will be approximately 25,593 SF with a finished floor elevation (FFE) of approximately 1028 Ft-MSL.

1.1.3 Maximum Loads

No structural loading information was provided as of the time of this exploration and report. However, based on our experience with similar projects, we estimate that the building will produce maximum column loads on the order of 150 kips with wall loads on the order of 3 to 5 kips per lineal foot (klf). We assume soil-supported ground floor loads (live loads) in the new building will not exceed 125 pounds per square foot (psf).

1.1.4 Floor Elevations / Site Grading

Based on the provided topographic information and the current proposed finished floor elevation of 1028 Ft-MSL, cuts and fills of up to 10 feet or less may be required to establish proposed finished grades.

1.2 SCOPE OF WORK

Lidl US, LLC engaged NOVA to provide geotechnical engineering consulting services for the Lidl - Norcross site in Norcross, GA. This report briefly discusses our understanding of the project, describes our exploratory procedures, and presents our findings, conclusions, and recommendations.

The primary objective of this study was to perform a geotechnical exploration within the area of the proposed construction and to assess these findings as they relate to geotechnical aspects of the planned site development. The authorized geotechnical engineering services included a site reconnaissance, a soil test boring and sampling program, engineering evaluation of the field and laboratory data, and the preparation of this report.

The services were performed as outlined in NOVA Proposal Number 003-20211025 dated February 11, 2021, and in general accordance with industry standards.

As authorized per the above referenced proposal, the completed geotechnical report was to include:

- ◆ A description of the site, fieldwork, laboratory testing, and general soil and groundwater conditions encountered, as well as a Boring Location Plan and individual Boring Records.
- ◆ Summary of percolation rate results obtained from the in-situ percolation tests.
- ◆ Discussion on potential earthwork-related issues indicated by the exploration, such as materials that would require difficult excavation techniques, unsuitable/deleterious materials, shallow groundwater table, etc.
- ◆ Recommendations for controlling groundwater and/or run-off during construction and the need for permanent dewatering systems based on the anticipated post construction groundwater levels.
- ◆ Suitability of on-site soils for re-use as structural fill and backfill. Additionally, the criteria for suitable fill materials and compaction requirements will be provided.
- ◆ Recommendations for foundation design and construction, including allowable bearing pressures and embedment depths.
- ◆ Estimate of total and differential settlements of foundations based on available structural loading data.
- ◆ Slab-on-grade construction considerations based on the geotechnical findings, including the need for a sub-slab vapor barrier or a capillary barrier and frost protection.
- ◆ Recommendations for an approximate coefficient of subgrade reaction.
- ◆ Recommendations for design and construction of slopes.

- ◆ Recommendations for lateral earth pressure coefficients for the design of below-grade walls.
- ◆ General recommendations for pavement design and construction.
- ◆ Recommended quality control measures (i.e. sampling, testing, and inspection requirements) for site grading and foundation construction.

The assessment of the presence of wetlands, floodplains, or water classified as State Waters of Georgia was beyond the scope of this study. Additionally, the assessment of site environmental conditions, including the detection of pollutants in the soil, rock, or groundwater, at the site was also beyond the scope of this geotechnical study. NOVA is pleased to provide these services if requested.

2.0 SITE DESCRIPTION

2.1 LOCATION AND LEGAL DESCRIPTION

The Subject Property consists of an approximately 3.01-acre tract of land located at 4990 Jimmy Carter Boulevard in Norcross, Gwinnett County, Georgia. According to the Gwinnett County Geographic Information System (GIS) database, the Subject Property is comprised of one (1) tax parcel identified by Parcel Identification Number 6169 064.

A Site Location Map and a Topographic Map depicting the location of the Subject Property and its surrounding topography are included in Appendix A (Figures 1 and 2).

2.2 SUBJECT PROPERTY AND VICINITY GENERAL CHARACTERISTICS

The vicinity of the Subject Property is generally developed with residential and commercial properties, and it is bordered by the following:

DIRECTION	LAND USE DESCRIPTION/OBSERVATIONS
NORTH	Restaurant
EAST	Jimmy Carter Boulevard, Commercial Stores
SOUTH	Shelby Drive NW, Restaurant and Gas Station
WEST	Residential Neighborhood

2.3 CURRENT USE OF THE PROPERTY

The Subject Property is currently developed as a warehouse event center with paved parking and drive areas. The northwest portion of the property is currently undeveloped and has medium to dense vegetation.

3.0 FIELD AND LABORATORY PROCEDURES

3.1 FIELD EXPLORATION

Boring locations were established in the field by NOVA personnel by utilizing a hand-held GPS unit and estimating distances and angles from existing site features. The approximate boring locations are shown on Figure 3 of Appendix A. Boring elevations were then interpolated from the provided site plan. Consequently, the referenced boring locations and elevations are approximate. If increased accuracy is desired by the client, NOVA recommends that the boring locations and elevations be surveyed.

Our field exploration was conducted on March 1, 2021 through March 3, 2021 and included:

- Six (6) SPT (Standard Penetration Test) soil test borings (B-3 through B-8) drilled to depths of up to 20 feet below the existing ground surface within the proposed footprint of the building extension and stormwater management pond.
- Two (2) hand auger/ DCP (Dynamic Cone Penetrometer) soil test borings (B-1 and B-2) excavated up to 10 feet below the existing ground surface within the footprint of the proposed building extension.
- Eleven (11) asphalt cores with base measurements and in-place CBR (California Bearing Ratio) correlation testing of the soil subgrade.
- One (1) percolation test performed in the area of the proposed stormwater pond.

Soil Test Borings: The soil test borings were performed using the guidelines of ASTM Designation D-1586, "Penetration Test and Split-Barrel Sampling of Soils". A hollow-stem auger was used to advance the borings. At regular intervals, soil samples were obtained with a standard 1.4-inch I.D., 2.0-inch O.D., split-tube sampler. The sampler was first seated six inches and then driven an additional foot with blows of a 140-pound hammer falling 30 inches. The number of hammer blows required to drive the sampler the final foot is designated the "Penetration Resistance". The penetration resistance, when properly interpreted, is an index to the soil strength and density. Representative portions of the soil samples, obtained from the sampler, were placed in glass jars and transported to our laboratory for further evaluation and laboratory testing.

Test Boring Records in Appendix B show the standard penetration test (SPT) resistances, or "N-values", and present the soil conditions encountered in the borings. These records represent our interpretation of the subsurface conditions based on the field exploration data, visual examination of the split-barrel samples, laboratory test data, and generally accepted geotechnical engineering practices. The stratification lines and depth designations represent approximate boundaries between various subsurface strata. Actual transitions between materials may be gradual.

Groundwater: The groundwater levels reported on the Test Boring Records represent measurements made at the completion of the soil test borings and 24 hours thereafter. The soil test borings were subsequently backfilled with the soil cuttings.

Pavement Study: Eleven (11) locations in the existing pavement were cored with a six-inch core barrel. The cores and any existing underlying base materials were measured to the nearest quarter inch. A Kessler Dynamic Cone Penetrometer (DCP) was used to estimate the in-place CBR of the existing subgrade soils supporting the existing pavement section in general accordance with ASTM D6951. Plots of the DCP Test Data collected at each test location are attached to this report. All the test holes were backfilled at the completion of testing and the holes were patched with high strength, quick-set concrete.

Percolation Tests: One (1) percolation test (P-1) was performed during our exploration. To perform the percolation test, a hand auger was utilized to drill bore hole P-1, in the vicinity of soil test boring B-7. The hole was drilled to a depth of approximately 4 feet, and approximately 4-6 inches of gravel was placed at the bottom of the borehole. The percolation test was performed in general accordance with standard test procedures adopted by the Georgia Department of Human Resources. The hole was filled with water to saturate the adjacent soil for a period of time dependent on the soil type, until percolation rates were consistent. The water level was then adjusted to approximately 6-12 inches above the gravel, and water level measurements were taken at recorded intervals until the level reached 2-4 inches above the gravel. Water was then added to re-establish the approximately 6-12 inch depth and measurements were continued. This was repeated until a constant rate of percolation was measured.

Following calculation of the percolation rate for the proposed test, the percolation rate was converted to an infiltration rate (inches/hour) via the Porchet method. The Porchet method accounts for the radius of the augered hole, and the average head over the final time interval. Furthermore, the infiltration rate considers only the downward progression of water into the soil as opposed to a percolation test, which also allows lateral flow through the soil.

3.2 LABORATORY TESTING

Split-barrel samples were returned to our testing laboratory, where they were classified using visual/manual methods in accordance with the Unified Soil Classification System (USCS) and ASTM designations. The descriptions presented in the boring logs should be considered approximate. It should be noted that all soil samples would be properly disposed of 30 days following the submittal of this NOVA subsurface exploration report unless you request otherwise.

Natural moisture testing was performed on one (1) bulk soil sample. The moisture content is the ratio expressed as a percentage of the weight of water in a given mass of soil to the weight of the solid particles. This test was conducted in general accordance with ASTM D 2216. The results of this test is included in the table below. Moisture tests were also performed on select SPT samples. These results are shown in the following table:

SPT Boring	Sample Depth (ft)	Natural Moisture Content (%)
B-4	3.5-5	20.3
B-4	6-7.5	25.3
B-4	8.5-10	13.8
B-5	3.5-5	23.3
B-5	6-7.5	28.7
B-5	8.5-10	21.6
B-8	3.5-5	25.2
B-8	6-7.5	22.5
B-8	8.5-10	15.8

A bulk soil sample collected from the vicinity of boring B-1 was returned to the laboratory for further testing, as described below:

3.2.1 Proctor Test

One (1) Standard Proctor compaction test was performed in accordance with ASTM D 698 – Standard Test Methods for Laboratory Compaction of Soil Using Standard Effort to determine the relationship between the soils' maximum dry unit weight and various moisture contents for use in controlling fill placement. The results are provided in Appendix C of this report, and in the table below.

3.2.2 Sieve Analysis

One (1) sieve analysis test was performed in this study. Sieve analysis consists of passing a soil sample through a series of standard sieve openings. The percentage of soil, by weight, passing the individual sieves is then recorded and generally presented in a graphical format. The percentage of fines passing through the No. 200 sieve is generally considered to represent the amount of silt and clay of the tested soil sample. The sieve analysis test was conducted in general accordance with ASTM Designation D 1140. Results are included on the Standard Proctor compaction test reports in Appendix C of this report, and in the table below.

3.2.3 Atterberg Limits

One (1) Atterberg Limits test was performed in this study. The Atterberg Limits are different descriptions of the moisture content of fine-grained soils as it transitions between a solid to a liquid-state. For classification purposes the two primary Atterberg Limits used are the plastic limit (PL) and the liquid limit (LL). The plastic index (PI) is also calculated for soil classification.

The plastic limit (PL) is the moisture content at which a soil transitions from being in a semisolid state to a plastic state. The liquid limit (LL) is defined as the moisture content at which a soil transitions from a plastic state to a liquid state. The tests were performed in this study in accordance with ASTM D4318. Results are included on the Liquid and Plastic Limits test reports in Appendix C of this report, and in the table below.

Bulk Sample	Natural Moisture Content (%)	Percent Passing Size 200 Sieve (%)	Optimum Moisture Content (%)	Maximum Dry Density (pcf)	Liquid Limit (LL)	Plastic Limit (PL)	USCS Classification
B-1	18.6	33.3	17.8	105.1	42	23	SC

4.0 SUBSURFACE CONDITIONS

4.1 GEOLOGY

The site is located in the Piedmont Geologic Region, a broad northeasterly trending province underlain by crystalline rocks up to 600 million years old. The Piedmont is bounded on the northwest by the Blue Ridge Range of the Appalachian Mountains, and on the southeast by the leading edge of Coastal Plain sediments, commonly referred to as the “Fall Line”. Numerous episodes of crystal deformation have produced varying degrees of metamorphism, folding, and shearing in the underlying rock. The resulting metamorphic rock types in this area of the Piedmont are predominantly a series of Precambrian age schists and gneisses, with scattered granitic or quartzite intrusions.

According to the "Geology of the Greater Atlanta Region" by McConnell and Abrams, 1984, the site is generally underlain by the **Southern Piedmont Province and Brevard Fault Zone, Clairmont Formation** (Higgins and Atkins, 1981), which consists of interlayered medium-grained biotite-plagioclase gneiss and fine to medium-grained hornblende-plagioclase amphibolite.

Residual soils in the region are primarily the product of in-situ chemical decomposition of the parent rock. The extent of the weathering is influenced by the mineral composition of the rock and defects such as fissures, faults and fractures. The residual profile can generally be divided into three zones:

- An upper zone near the ground surface consisting of red clays and clayey silts which have undergone the most advanced weathering,
- An intermediate zone of less weathered micaceous sandy silts and silty sands, frequently described as “saprolite”, whose mineralogy, texture and banded appearance reflects the structure of the original rock, and
- A transitional zone between soil and rock termed partially weathered rock (PWR). Partially weathered rock is defined locally by standard penetration resistances exceeding 100 blows per foot.

The boundaries between zones of soil, partially weathered rock, and bedrock are erratic and poorly defined. Weathering is often more advanced next to fractures and joints that transmit water, and in mineral bands that are more susceptible to decomposition. Boulders and rock lenses are sometimes encountered within the overlying PWR or soil matrix. Consequently, significant fluctuations in depths to materials requiring difficult excavation techniques may occur over short horizontal distances.

4.2 SOIL AND ROCK CONDITIONS

The following paragraphs provide generalized descriptions of the subsurface profiles and soil conditions encountered by the borings conducted during this study.

The Test Boring Records in Appendix B should be reviewed to provide more detailed descriptions of the subsurface conditions encountered at each boring location. These records represent our interpretation of the subsurface conditions based on the field logs and visual observations of samples by an engineer. The lines designating the interface between various strata on the Boring Logs represent the approximate interface locations and elevation. The actual transition between strata may be gradual. Groundwater levels shown on the Boring Logs represent the conditions at the time of drilling. It should be understood that soil conditions may vary between boring locations.

4.2.1 Surface Materials

Topsoil: Topsoil was encountered at each boring location (except for B-6 and B-7), ranging in thickness from approximately 1 to 4 inches. Please note that topsoil thicknesses are frequently erratic and thicker zones of topsoil should be anticipated, most notably within the heavily wooded portions of the site.

Mulch/Woodchips: Approximately one (1) foot of mulch and/or woodchips were encountered in the area of soil test borings B-6 and B-7. It appears that this layer was created from the mulching of trees and vegetation previously present in the area of the proposed stormwater management pond.

4.2.2 Fill Soils

Fill soils were encountered in all borings. The fill generally consisted of very loose to medium dense clayey and silty medium to fine SAND. Standard penetration resistance values ranged from were 0 bpf to 11 bpf. The table below shows the depths of fill encountered, and the approximate elevation of the bottom of fill.

BORING	DEPTH (feet)	APPROXIMATE ELEVATION (feet-MSL)
B-1*	5.0+	<1031
B-2*	10.0+	<1026
B-3	12.0	1019
B-4	12.0	1018
B-5	12.0	1018
B-6	12.0	1014
B-7*	15.0+	<1013

BORING	DEPTH (feet)	APPROXIMATE ELEVATION (feet-MSL)
B-8	12.0	1017

* Fill in these locations likely extends deeper than the depth at which the boring refused or terminated.

4.2.3 Residual Soils

Residual soils were encountered in all borings that extended beneath the near surface fill materials. The residuum generally consisted of loose red-brown silty medium to fine SAND. Standard penetration resistance values ranged from 6 to 10 bpf.

4.2.4 Partially Weathered Rock

Partially weathered rock (PWR) is a transitional material between soil and the underlying parent rock that is defined locally as materials that exhibit a standard penetration resistance exceeding 100 bpf. PWR was not encountered in any borings to the depths explored.

4.2.5 Auger Refusal Materials

Auger refusal materials are any very hard or very dense material, frequently boulders or the upper surface of bedrock, which cannot be penetrated by a power auger.

Auger refusal was not encountered in any of our soil test borings but was encountered in hand auger boring B-2 at a depth of 5 feet below existing grade (approximately 1031 ft-MSL). It should be noted that unlike power auger refusals referenced above, hand auger refusals are often caused by small obstructions embedded in the soil such as small rocks and roots. Therefore, this auger refusal is not necessarily indicative of very hard or very dense material. Auger refusal data is shown in the Boring Logs in Appendix B of this report.

4.3 GROUNDWATER CONDITIONS

4.3.1 General

Groundwater in the Piedmont typically occurs as an unconfined or semi-confined aquifer condition. Recharge is provided by the infiltration of rainfall and surface water through the soil overburden. More permeable zones in the soil matrix, as well as fractures, joints and discontinuities in the underlying bedrock can affect

groundwater conditions. The groundwater table in the Piedmont is expected to be a subdued replica of the original surface topography.

Groundwater levels vary with changes in season and rainfall, construction activity, surface water runoff, and other site-specific factors. Groundwater levels in the Norcross area are typically lowest in the late summer-early fall and highest in the late winter-early spring, with annual groundwater fluctuations of 4 to 8 feet; consequently, the water table may vary at times.

4.3.2 Soil Test Boring Groundwater Conditions

Groundwater was encountered in soil test boring B-7 at a depth of approximately seven (7) feet below existing grade (approximately 1021 Ft-MSL).

4.4 PAVEMENT ANALYSIS

4.4.1 Visual Assessment

The existing driveways and parking lots were visually assessed to determine the nature and extent of existing surface defects. Extensive severe block cracking and fatigue cracking (colloquially known as “alligator cracking”) were observed throughout the pavement area, along with large deteriorating patches, potholes, and joint raveling. The existing pavements were generally observed to be in poor condition and are beyond their useful service life.

4.4.2 Pavement Coring and DCP/CBR Testing

Eleven (11) locations in the existing pavement were cored with a six-inch core barrel. The cores and any existing underlying base materials were measured to the nearest quarter inch. As shown in the table below, asphalt thicknesses ranged from approximately 1.75 to 3.25 inches. The asphalt layer was generally underlain by a graded aggregate base (GAB) layer ranging in thickness from 3.5 to 7.5 inches (core location C-10 had no GAB beneath the asphalt.)

The cores obtained indicated that cracks in the pavement extended the full thickness of the asphalt. This allows for intrusion of stormwater runoff, which can lead to weakening of the base and subgrade, which will in turn accelerate deterioration of the pavement section.

A Kessler Dynamic Cone Penetrometer (DCP) was used to estimate the in-place CBR of the existing subgrade soils supporting the existing pavement section in general accordance with ASTM D6951. Plots of the DCP Test Data collected at

each test location are attached to this report and summarized in the table below. All the test holes were backfilled at the completion of testing and the holes were patched with high strength, quick-set concrete.

CORE LOCATION	ASPHALT THICKNESS (Inches)	GAB THICKNESS (Inches)	ESTIMATED IN-PLACE CBR VALUE
C-1	2 ¼	5 ¾	4
C-2	2 ¼	6 ½	8
C-3	2 ¾	6 ¼	10
C-4	3 ⅛	6 ½	3
C-5	2 ¾	3 ¾	9
C-6	3 ⅛	6	4
C-7	2 ¾	5 ¼	8
C-8	3 ¼	7 ½	2
C-9	2 ½	6	4
C-10	2 ½	0	10
C-11	1 ¾	3 ½	6

5.0 CONCLUSIONS AND RECOMMENDATIONS

The following conclusions and recommendations are based on our understanding of the proposed construction, site observations, our evaluation and interpretation of the field and laboratory data obtained during this exploration, our experience with similar subsurface conditions, and generally accepted geotechnical engineering principles and practices.

Subsurface conditions in unexplored locations or at other times may vary from those encountered at specific boring locations. If such variations are noted during construction, or if project development plans are changed, we request the opportunity to review the changes and amend our recommendations, if necessary.

As previously noted, boring locations were established by estimating distances and angles from site landmarks. If increased accuracy is desired by the client, we recommend that the boring locations and elevations be surveyed.

5.1 SITE PREPARATION

5.1.1 General

General: Several structures that will require demolition currently occupy the site. Prior to proceeding with construction, all structures, vegetation, root systems, topsoil, and other deleterious non-soil materials should be stripped from proposed construction areas. Clean topsoil may be stockpiled and subsequently re-used in landscaped areas. Debris-laden materials should be excavated, transported, and disposed of off-site in accordance with appropriate solid waste rules and regulations. All existing utility locations should be reviewed to assess their impact on the proposed construction and relocated/grouted in-place as appropriate.

After clearing and stripping, areas, which are at grade or will receive fill should be carefully evaluated by a NOVA geotechnical engineer. The engineer will require proofrolling of the subgrade with multiple passes of a 20 to 30 ton loaded truck, or other vehicle of similar size and weight.

The purpose of the proofrolling is to locate soft, weak, or excessively wet fill or residual soils present at the time of construction. Unstable materials observed during the evaluation and proof-rolling operations should be undercut and replaced with structural fill or stabilized in-place by scarifying and re-densifying.

In the event that low consistency and/or debris laden fill materials are encountered during construction, typical recommendations would include undercutting and backfilling with structural fill and/or stabilizing in-place with

fabric, stone, and/or other remedial techniques. Actual remedial recommendations can best be determined by the geotechnical engineer in the field at the time of construction.

The site should be graded during construction such that positive drainage is maintained away from the construction areas, to prevent ponding of storm water on the site during and shortly following significant rain events. The construction areas should also be sealed and crowned with a smooth roller to minimize ponding water from storm events at the end of each day of work. The fine-grained soils encountered during our exploration are weather sensitive and will be susceptible to loss of strength and density if exposed to freeze/thaw and or wetting/drying cycles. Care must be exercised by the grading contractor to protect subgrades and new fills from inclement weather.

5.1.2 Existing / Old Fill

Previously placed fill materials were encountered during this exploration. Based on our experience, we anticipate fill materials likely exist at other locations between our borings. If low consistency and/or debris-laden fill materials are encountered during construction, typical recommendations would include undercutting and backfilling with structural fill and/or stabilizing in-place with fabric, stone, and/or other remedial techniques. Actual remedial recommendations can best be determined by the geotechnical engineer in the field at the time of construction.

Poor consistency fill materials were encountered extending to depths of up to 10 feet in borings B-3, B-4, B-6, B-7 and B-8. These soils are not suitable to support the proposed building or pavements in their current condition. These materials will require stabilization or undercutting and replacement with structural fill to support structures or pavements.

Additionally, natural moisture content testing indicates that the in-place fill soils are above optimum water content. These soils would need to be aerated and dried to within 3 percent of optimum moisture in order to be reused as structural fill.

The consistency of the existing fills generally improves approaching the existing building and may not require extensive remediation within the footprint of the proposed building addition. Should low consistency materials be encountered immediately adjacent to the existing building, care should be exercised to not undermine the existing building foundations during remediation. The use of helical piers to support the existing foundations may be required depending upon the required excavation depths.

5.2 FILL PLACEMENT

5.2.1 Fill Suitability

Fill materials should be low plasticity soil (Plasticity Index less than 30), free of non-soil materials and rock fragments larger than 3 inches in any one dimension. Based on visual examination, the existing residual soils and much of the existing fill, which does not contain appreciable amounts of debris, rock organics or other deleterious materials, encountered during this exploration generally appear suitable for re-use as structural fill. Prior to construction, bulk samples of the proposed fill materials should be laboratory-tested to confirm their suitability.

Organic and/or debris-laden material is not suitable for re-use as structural fill. Topsoil, mulch, and similar organic materials can be wasted in architectural areas. Debris-laden materials should be excavated, transported, and disposed of off-site in accordance with appropriate solid waste rules and regulations.

5.2.2 Soil Compaction

Fill should be placed in thin, horizontal loose lifts (maximum 8-inch) and compacted to at least 95 percent of the standard Proctor maximum dry density (ASTM D 698). In confined areas, such as utility trenches or behind retaining walls, portable compaction equipment and thinner fill lifts (3 to 4 inches) may be necessary. Fill materials used in structural areas should have a target maximum dry density of at least 95 pounds per cubic foot (pcf). If lighter weight fill materials are used, the NOVA geotechnical engineer should be consulted to assess the impact on design recommendations.

Soil moisture content should be maintained within 3 percent of the optimum moisture content. We recommend that the grading contractor have equipment on site during earthwork for both drying and wetting fill soils. Moisture control may be difficult during rainy weather. If soils are excavated near or below the groundwater table, they will require significant efforts to achieve acceptable moisture contents prior to re-use as fill.

Filling operations should be observed by a NOVA soils technician, who can confirm suitability of material used and uniformity and appropriateness of compaction efforts. He/she can also document compliance with the specifications by performing field density tests using nuclear or sand cone testing methods (ASTM D 6938, or D 1556, respectively). One test per 400 cubic yards and every 2 feet of placed fill is recommended, with test locations well distributed throughout the fill mass. When filling in small areas, at least one test per day per area should be performed.

5.3 GROUNDWATER CONTROL

5.3.1 General

During the current study, groundwater was encountered at soil test boring B-7 at a depth of seven (7) feet below existing grades (approximately 1021 Ft-MSL).

Based on the estimated finished floor elevation of approximately 1030 feet-MSL, we do not anticipate significant groundwater control problems during mass grading in the building and surrounding areas of the site.

As previously noted, groundwater levels are subject to seasonal, climatic and other variations and may be different at other times and locations. The extent and nature of any dewatering required during construction will be dependent on the actual groundwater conditions prevalent at the time of construction and the effectiveness of construction drainage to prevent run-off into open excavations.

5.4 SLOPES

All new fill placed should be properly benched into the existing slopes and should be compacted in accordance with the requirements outlined in Section 5.2. It is good practice to over-build slopes and then back cut the slopes to the proper grade.

Slope stability analysis using laboratory shear strength data was beyond the scope of this study. However, based on our experience with similar subsurface conditions and construction, permanent slopes no steeper than 2.0(H): 1.0(V) should be stable long term, if limited in height to 25 feet, and are not inundated or subjected to rapid draw-down conditions, or subjected to groundwater seepage.

Adjacent to buildings, a top of slope set-back of 10 feet is recommended. In pavement areas, a minimum top of slope setback of 5 feet is acceptable.

Temporary slopes should be no steeper than OSHA guidelines. During construction, temporary slopes should be regularly inspected for signs of movement or unsafe condition. Soil slopes should be covered for protection from rain, and surface run-off should be diverted away from the slopes. For erosion protection, a protective cover of grass or other vegetation should be established on permanent soil slopes as soon as possible.

5.5 FOUNDATIONS

5.5.1 General

Detailed structural loading information was not provided. However, based on our experience with similar projects, we anticipate column and wall loads will not exceed 150 kips and 5 to 7 kips per linear foot, respectively.

5.5.2 Shallow Foundations

Design: After the recommended site and subgrade preparation and fill placement, we recommend that the proposed building be supported by conventional shallow foundations. Foundations bearing on undisturbed residual soils and/or compacted structural fill may be designed for a maximum allowable bearing pressure of 2,000 pounds per square foot (psf).

Previously placed fill materials were encountered on the site. In the event that low consistency or debris-laden fill materials are present in foundation excavations, undercutting and backfilling with crushed stone or structural fill or redesigning for a reduced bearing pressure, may be required.

As discussed in section 5.1.2, remediation of some of the existing fill materials will be required. Should low consistency materials be encountered immediately adjacent to the existing building, care should be exercised to not undermine the existing building foundations during remediation. The use of helical piers to support the existing foundations may be required depending upon the required excavation depths.

We recommend minimum foundation widths of 24 inches for ease of construction and to reduce the possibility of localized shear failures. Exterior foundation bottoms should be at least 18 inches below exterior grades for protection against frost damage.

Settlement: Settlements for spread foundations bearing on the higher consistency residual materials were assessed using SPT values to estimate elastic modulus, based on published correlations and previous NOVA experience. We note that the settlements presented are based on random field data and an assumed subsoil profile. Conditions may be better or worse in other areas, however, we believe the estimated settlements are reasonably conservative. The time rate of settlement was estimated based on NOVA's experience with similar data and soil profiles/based upon the results of the consolidation testing.

Based on anticipated maximum column loads of 150 kips, soil bearing capacities and the presumed foundation elevations as discussed above, we expect primary total settlement beneath individual foundations to be less than 1 inch.

The amount of differential settlement is difficult to predict because the subsurface and foundation loading conditions can vary considerably across the site. However, we anticipate differential settlement between adjacent foundations to be less than ½ inch. The final deflected shape of the structure will be dependent on actual foundation locations and loading.

Please note that if actual design column loads differ significantly from the provided loads, NOVA should be notified immediately in order to verify that the foundation recommendations and anticipated settlements are still appropriate for the actual design loads.

To reduce the differential settlement if lower consistency materials are encountered, a lower bearing capacity should be used or the foundations should be extended to more competent materials. In addition, foundation subgrades which are excavated into PWR/rock may need to be slightly undercut with controlled structural fill placed between the PWR/rock and the bottom of the foundation to produce some settlement of the foundation, thus reducing differential settlements with nearby foundations bearing on less dense material. We anticipate that timely communication between the geotechnical engineer and the structural engineer, as well as other design and construction team members, will be required.

Construction: Foundation excavations should be evaluated by the NOVA geotechnical engineer prior to reinforcing steel placement to observe foundation subgrade preparation and confirm bearing pressure capacity.

Foundation excavations should be level and free of debris, ponded water, mud, and loose, frozen, or water-softened soils. Concrete should be placed as soon as is practical after the foundation is excavated and the subgrade evaluated. Foundation concrete should not be placed on frozen or saturated soil. If a foundation excavation remains open overnight, or if rain or snow is imminent, a 3 to 4-inch thick "mud mat" of lean concrete should be placed in the bottom of the excavation to protect the bearing soils until reinforcing steel and concrete can be placed.

5.6 SLAB-ON-GRADE

5.6.1 General

The conditions exposed at subgrade levels will vary across the site and may include structural fill and/or residual soils. Slabs-on-grade may be adequately supported on these subgrade conditions subject to the recommendations in this report. Slabs-on-grade should be jointed around columns and along walls to reduce cracking due to differential movement. A 4-inch layer of crushed stone may be placed beneath the building slabs to reduce non-uniform support conditions.

An underdrain system is not necessary beneath the slabs, but an impermeable vapor barrier is recommended beneath finished spaces to reduce dampness.

Once grading is completed, the subgrade is usually exposed to adverse construction activities and weather conditions during the period of sub-slab utility installation. The subgrade should be well-drained to prevent the accumulation of water. If the exposed subgrade becomes saturated or frozen, the geotechnical engineer should be consulted.

After utilities have been installed and backfilled, a final subgrade evaluation should be performed by the geotechnical engineer immediately prior to slab-on-grade placement. If practical, proofrolling may be used to redensify the surface and to detect any soil that has become excessively wet or otherwise loosened.

5.6.2 Subgrade Modulus

A coefficient of subgrade reaction (k) of 125 pci (psi per inch) may be used for conventional slab design where slabs bear upon subgrades prepared in accordance with previous recommendations.

Please note that this magnitude of k is intended to reflect the elastic response of soil beneath a typical floor slab under light loads with a small load contact area often measured in square inches, such as loads from forklifts, automobile/truck traffic or lightly loaded storage racks. The recommended coefficient of subgrade reaction (k) of 125 pci is not applicable for heavy slab loads caused by bulk storage or tall storage racks, or for mat foundation design. Several design methods are applicable for conventional slab design. We have assumed that the slab designer will utilize the methods discussed in the American Concrete Institute (ACI) Committee 360 report, “*Guide to Design of Slabs-on-Ground*, (ACI 360R-10), specifically the Portland Cement Association (PCA), or the Wire Reinforcement Institute (WRI) slab thickness design methods.

5.7 BELOW GRADE WALLS

The magnitude and distribution of earth pressures against below grade walls depends on the deformation condition (rotation) of the wall, soil properties and water conditions. When the soil behind the wall is prevented from lateral strain, the resulting force is known as the at-rest earth pressure (K_0). If the retaining structure moves away from the soil mass, the earth pressure decreases with the increasing lateral expansion until a minimum pressure, known as the active earth pressure (K_A), is reached. If the wall is forced into the soil mass, the earth pressure increases until a maximum pressure, known as the passive earth pressure (K_P), is obtained.

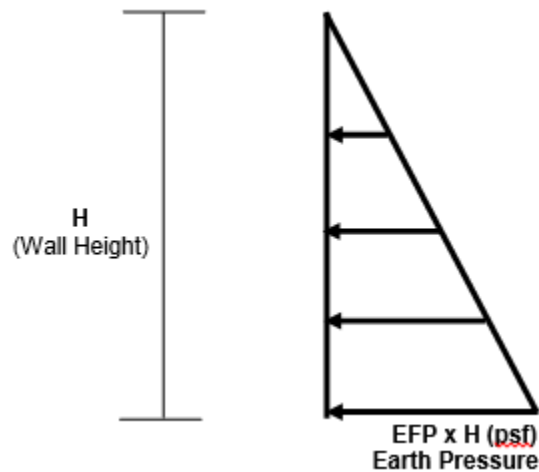
Free-standing retaining walls are usually designed for active earth pressures. Rigid basement walls are typically designed for at-rest earth pressures. If basement walls will be backfilled before they are braced by the floor slabs, they should also be designed to withstand active earth pressures as self-supporting cantilever walls. However, the earth pressures must be compatible with the wall rotation, which is limited by the wall rigidity, foundation support conditions and connections to adjoining structures. If active earth pressure development requires horizontal wall movements that cannot occur, or which are architecturally undesirable, walls should be designed for an intermediate pressure based on restraint conditions.

Laboratory analysis to determine actual soil shear strength properties was beyond the authorized scope of services. Based on our experience with similar soils and construction, we have provided the earth pressure estimates shown in the following table.

Earth Pressure Condition	Earth Pressure Coefficient	Equivalent Fluid Pressure (pcf)	
		Above Water Table	Below Water Table
Soil Backfill			
Active (K_a)	0.33	40	80
At-Rest (K_o)	0.50	60	89
Passive (K_p)	3.00	150*	TBD**
#57 Stone Backfill			
Active (K_a)	0.29	35	75
At-Rest (K_o)	0.46	55	84
Passive (K_p)	3.40	400*	TBD**

* Passive earth pressure is frequently used in retaining wall design to resist active earth pressures. Wall movements required to develop full passive earth pressures are significantly greater than movements necessary for active earth pressures. Consequently, this passive pressure value has been reduced by at least 50% for wall design

** Passive earth pressure for submerged wall design shall be determined on a case-by-case basis.



We recommend a value of 0.35 as the coefficient of friction (sliding resistance) between wall foundations and the underlying residual or fill soils. A coefficient of friction of 0.45 is recommended for foundations bearing on PWR. A coefficient of friction of 0.5 is recommended for foundations bearing on rock. These design values do not contain a safety factor.

Our lateral earth pressure recommendations assume that:

- The ground surface adjacent to the wall is level,
- Residual soils will be reused for wall backfill, compacted between 95% to 98% of the standard proctor maximum dry density,
- Soil backfill weight is a maximum of 120 pcf,
- Heavy construction equipment does not operate within 5 feet of the walls,
- A constantly functioning drainage system is installed between the wall and the soil backfill to prevent hydrostatic pressures from acting on the wall,
- Foundations or other significant surcharge loads are located outside the wall a distance at least equal to the wall height,
- For active earth pressure, wall must rotate about base, with top lateral movements of about $0.002 H$ to $0.004 H$, where H is wall height,
- For passive earth pressure to develop, wall must move horizontally to mobilize resistance.

5.8 PAVEMENT SECTIONS

The following pavement section recommendations is based on traffic loading conditions identified in the Lidl Design Development guidelines as follows:

- Design life of 20 years based on a minimum ESAL (18 kip) load as shown below, terminal serviceability = 2.0, reliability = 85%, initial serviceability =

4.2, standard deviation = 0.45 for flexible pavements and 0.35 for rigid pavements.

- Minimum Standard Duty 20-Year Design Life ESAL's = 35,000
- Minimum Heavy Duty 20-Year Design Life ESAL's = 75,000
- Paving shall be designed using the "AASHTO Guide for Design of Pavement Structures 1993".

The existing pavements across the site are in poor condition and are beyond their useful service life. We understand the pavements will be reconstructed during site development and the following are design recommendations for the new pavements.

5.8.1 Flexible Pavement

Site-specific traffic loading data was not available for this exploration. However, based on subsurface conditions encountered at this site, the recommended site preparation, and an estimated CBR of 4, our typical recommended flexible pavement design for a retail facility is shown in the following table:

Pavement Section	Standard Duty *	Heavy Duty **
Asphaltic Surface Course (9.5 mm SuperPave, GDOT approved mix)	2 inches	2 inches
Asphaltic Base Course (19 mm SuperPave, GDOT approved mix)	N/A	2 inches
Graded Aggregate Base (GAB) or Concrete Base Course (from an approved GDOT source)	8 inches	8 inches
Stabilized Subgrade (compacted to a minimum 98% of the standard Proctor maximum dry density)	12 inches	12 inches

* Standard Duty – Driveways and parking lots restricted to automobile traffic

** Heavy Duty – Driveways and parking lots subject to automobile and truck traffic

We recommend a minimum compaction of 98 percent of the maximum dry density for the Graded Aggregate Base (GAB) as determined by the modified Proctor compaction test (ASTM D 1557, Method C). The crushed stone should conform to applicable sections of the current GDOT Standard Specifications. All asphalt material and paving operations should meet applicable specifications of the Asphalt Institute and GDOT. A NOVA technician should observe placement and perform density testing of the base course material and asphalt.

5.8.2 Rigid Pavement

A rigid pavement section is recommended in areas where heavy truck traffic, excessive braking, sharp wheel turning and/or point loads, like dumpsters and loading docks are planned. Based on the subsurface conditions at the site, the recommended site preparation, assumed low volume traffic loads and an estimated subgrade modulus (k) of 125 psi/inch for traffic or wheel loading, our recommended rigid pavement design is as follows:

Pavement Section	Standard Duty *	Heavy Duty **
GDOT approved air-entrained concrete mix	5 inches	7 inches
Graded Aggregate Base (GAB) or Concrete Base Course (from an approved GDOT source)	4 inches	4 inches
Control Joint Spacing (maximum)	10 feet X 10 feet	12 feet X 12 feet
Saw-Cut Depth (minimum)	1½ inches	2 inches

* Standard Duty – Driveways and parking lots restricted to automobile traffic

** Heavy Duty – Driveways and parking lots subject to automobile and truck traffic

All concrete materials and placement should conform to applicable GDOT specifications. We recommend that a non-woven geotextile (about 3 feet wide) be placed beneath the construction joints to prevent upward "pumping" movement of soil fines through the joints.

We recommend using concrete with a minimum compressive strength of 4,000 psi and a minimum 28-day flexural strength (modulus of rupture) of at least 600 pounds per square inch, based on 3-point loading of concrete beam test samples. Layout of the saw-cut control joints should form square panels, and the depth of saw-cut joint should be approximately $\frac{1}{4}$ of the concrete slab thickness. The joints should be sawed within six (6) hours of concrete placement or as soon as the concrete has developed sufficient strength to support workers and equipment.

We recommend allowing NOVA to review and comment on the final concrete pavement design, including section and joint details (type of joints, joint spacing, etc.), prior to the start of construction. For further details on concrete pavement

construction, please reference the “Building Quality Concrete Parking Areas”, published by the Portland Cement Association.

Please note that the recommended pavement section is based on assumed post-construction traffic loading. If the pavement is to be constructed and utilized by construction traffic, the above pavement section will likely prove insufficient for heavy truck traffic, such as concrete trucks or tractor-trailers used for construction delivery. Unexpected distress, reduced pavement life and /or premature failure of the pavement section could result if subjected to heavy construction traffic and the owner should be made aware of this risk. If the assumed traffic loading stated herein is not correct, NOVA should review actual pavement loading conditions to determine if revisions to these recommendations are warranted.

5.9 SEISMIC SOIL SITE CLASS

In accordance with Section 1613.3.2 of the 2018 International Building Code (IBC), the seismic Site Class was estimated using the standard penetration resistance values obtained from the soil test borings performed during this study. Based upon this analysis, and our knowledge of general subsurface conditions in the area, we believe the soil profiles associated with a Site Class “D” are generally appropriate for this site.

6.0 CONSTRUCTION OBSERVATIONS

6.1 SUBGRADE

Once site grading is completed, the subgrade may be exposed to adverse construction activities and weather conditions. The subgrade should be well-drained to prevent the accumulation of water. If the exposed subgrade becomes saturated or frozen, the NOVA geotechnical engineer should be consulted.

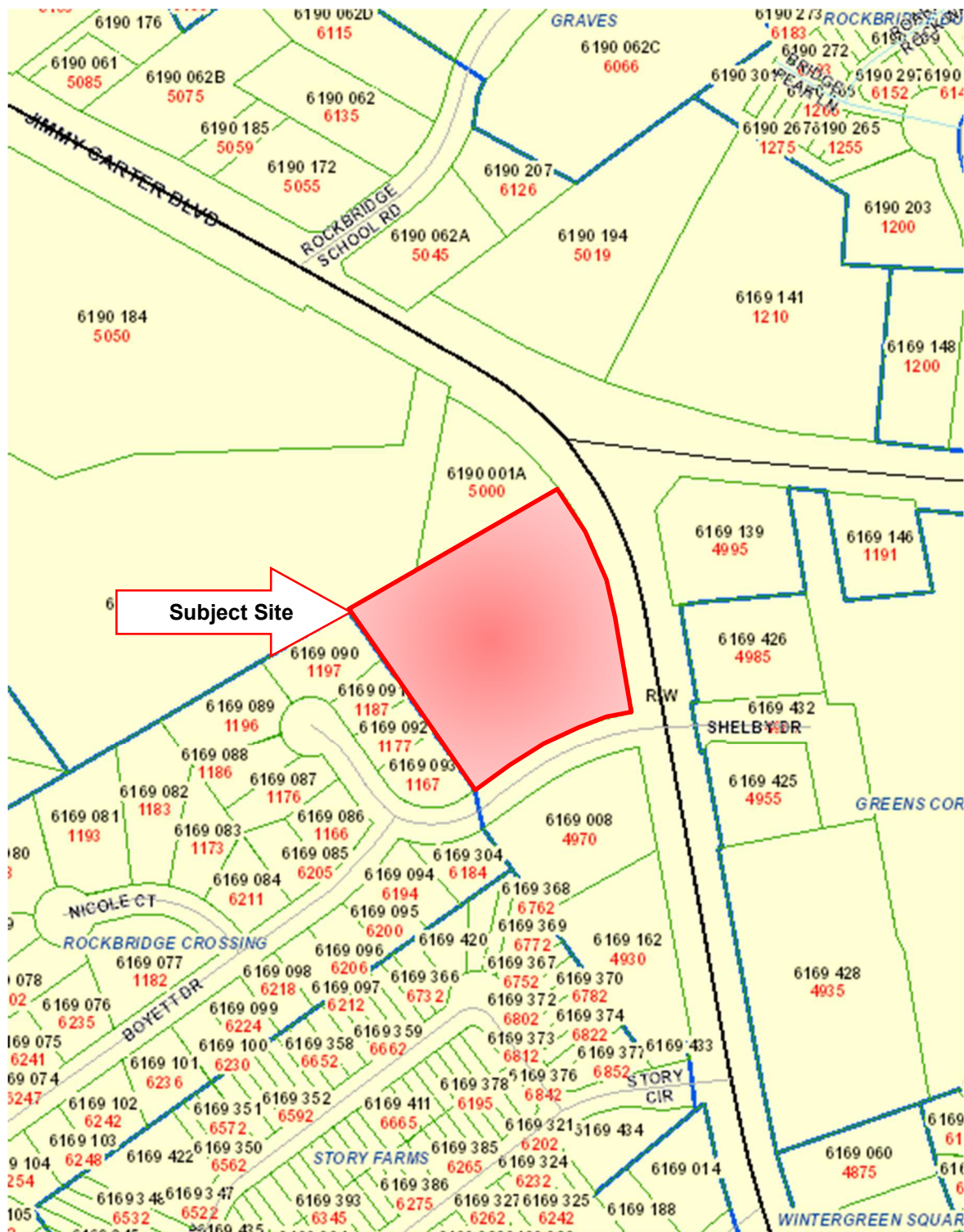
A final subgrade evaluation should be performed by the NOVA geotechnical engineer immediately prior to pavements. If practical, proof-rolling may be used to re-densify the surface and to detect any soil, which has become excessively wet or otherwise loosened.

6.2 SHALLOW FOUNDATIONS

Foundation excavations should be level and free of debris, ponded water, mud, and loose, frozen or water-softened soils. All foundation excavations should be evaluated by the NOVA geotechnical engineer prior to reinforcing steel placement to observe foundation subgrade preparation and confirm bearing pressure capacity. Due to variable site subsurface and construction conditions, some adjustments in isolated foundation bearing pressures, depth of foundations or undercutting and replacement with controlled structural fill may be necessary.

APPENDIX A

Figures and Maps



Approximate Property Boundary

FIGURE 1
SITE LOCATION

Source: Gwinnett County GIS Parcel Viewer, Accessed 2-27-2021
Scale: Not To Scale



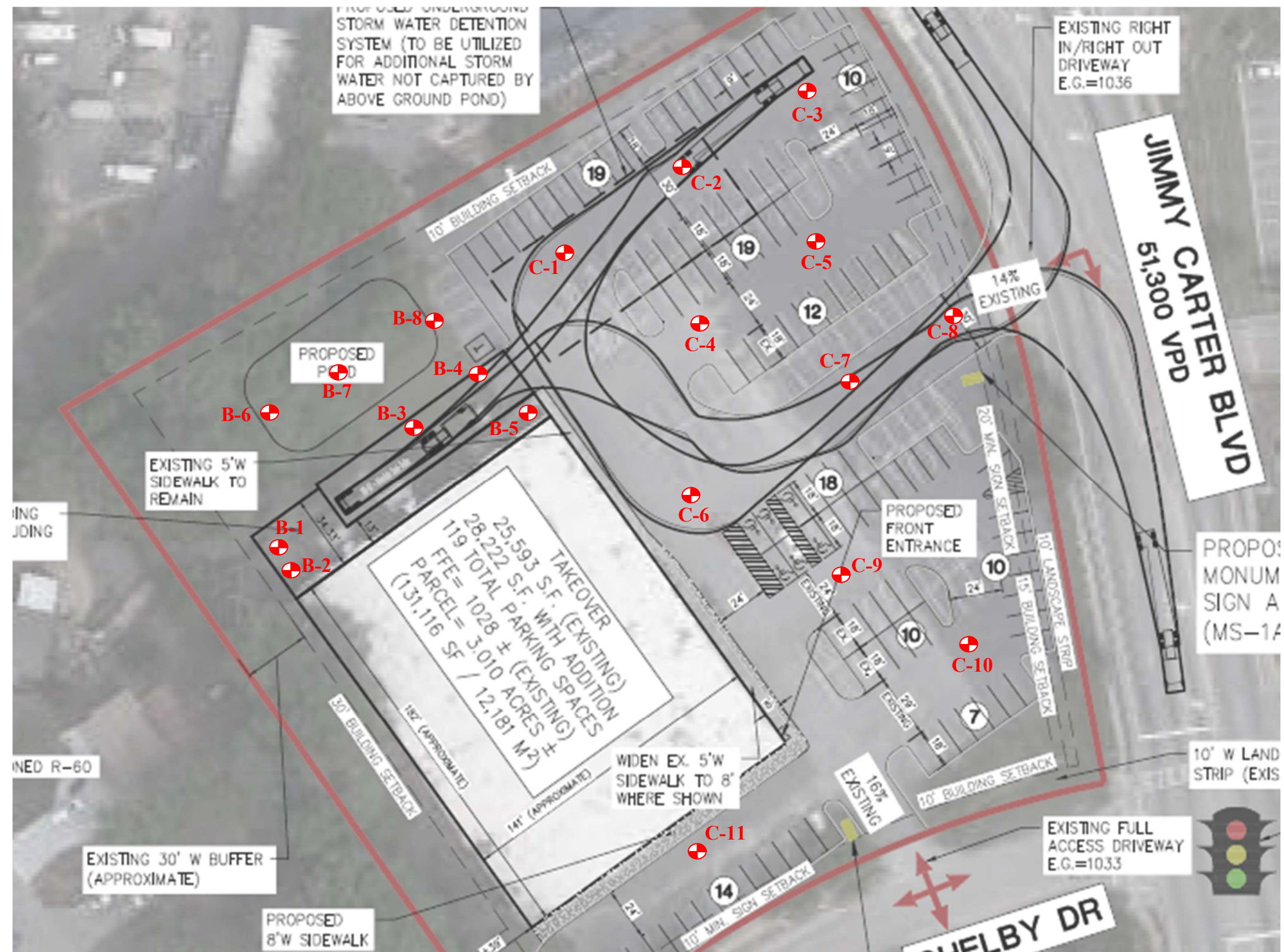
Lidl US, LLC
Lidl - Norcross
4990 Jimmy Carter Boulevard
Norcross, Gwinnett County, Georgia
NOVA Project Number 10103-2021006



FIGURE 2
TOPOGRAPHIC MAP
 Source: USGS Topoview
 Norcross, GA, 2020
 Scale: Not To Scale



Lidl US, LLC
 Lidl - Norcross
 4990 Jimmy Carter Boulevard
 Norcross, Gwinnett County, Georgia
 NOVA Project Number 10103-2021006



● APPROXIMATE LOCATION OF NOVA SOIL TEST BORINGS

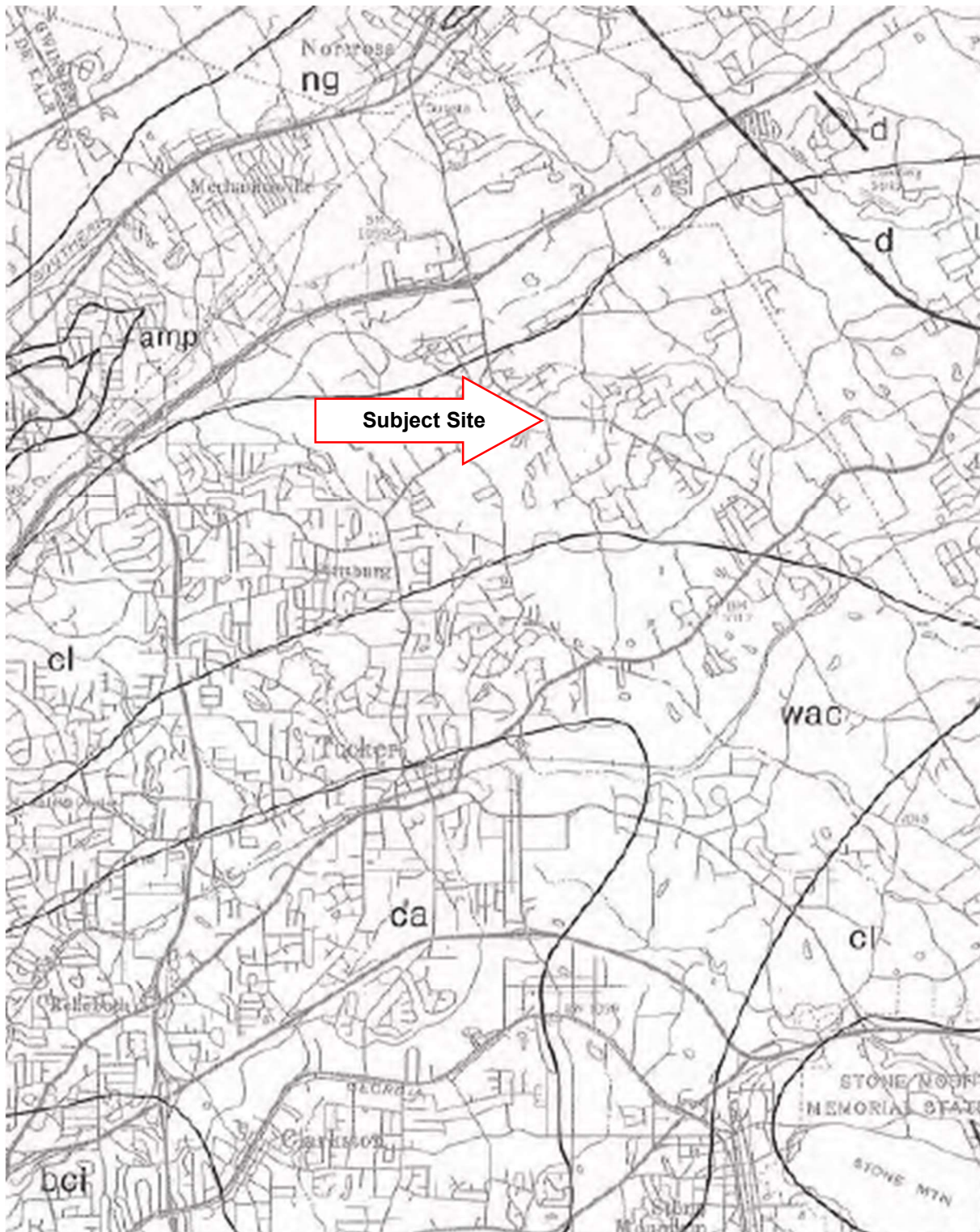
FIGURE 3
BORING LOCATION PLAN

SCALE: Not To Scale

SOURCE: Overall Site Plan, Drawn by CHA, dated 11-10-2020



Lidl US, LLC
Lidl - Norcross
4990 Jimmy Carter Boulevard
Norcross, Gwinnett County, Georgia
NOVA Project Number 10103-2021006



Southern Piedmont Province and Brevard Fault Zone, Clairmont Formation (Higgins and Atkins, 1981): interlayered medium-grained biotite-plagioclase gneiss and fine to medium-grained hornblende-plagioclase amphibolite. (cl)

FIGURE 4 REGIONAL GEOLOGY

SOURCE: Geology of The Greater Atlanta Region, GA
DNR/EPD/GGS, McConnell and Abrams, Bulletin 96,
1984

Scale: Not To Scale



Lidl US, LLC
Lidl - Norcross
4990 Jimmy Carter Boulevard
Norcross, Gwinnett County, Georgia
NOVA Project Number 10103-2021006

APPENDIX B

Subsurface Data

KEY TO SYMBOLS AND CLASSIFICATIONS

DRILLING SYMBOLS

	Split Spoon Sample
	Undisturbed Sample (UD)
	Standard Penetration Resistance (ASTM D1586-67)
	Water Table at least 24 Hours after Drilling
	Water Table 1 Hour or less after Drilling
100/2"	Number of Blows (100) to Drive the Spoon a Number of Inches (2)
NX, NQ	Core Barrel Sizes: 2½- and 2-Inch Diameter Rock Core, Respectively
REC	Percentage of Rock Core Recovered
RQD	Rock Quality Designation – Percentage of Recovered Core Segments 4 or more Inches Long
	Loss of Drilling Water
MC	Moisture Content Test Performed

CORRELATION OF PENETRATION RESISTANCE WITH RELATIVE DENSITY AND CONSISTENCY

	<u>Number of Blows, "N"</u>	<u>Approximate Relative Density</u>
	0 – 4	Very Loose
	5 – 10	Loose
SANDS	11 – 30	Medium Dense
	31 – 50	Dense
	Over 50	Very Dense
	<u>Number of Blows, "N"</u>	<u>Approximate Consistency</u>
	0 – 2	Very Soft
	3 – 4	Soft
SILTS	5 – 8	Firm
and	9 – 15	Stiff
CLAYS	16 – 30	Very Stiff
	31 – 50	Hard
	Over 50	Very Hard

DRILLING PROCEDURES

Soil sampling and standard penetration testing performed in accordance with ASTM D1586-67. The standard penetration resistance is the number of blows of a 140 pound hammer falling 30 inches to drive a 2-inch O.D., 1½-inch I.D. split spoon sampler one foot. Core drilling performed in accordance with ASTM D2113-08. The undisturbed sampling procedure is described by ASTM D1587-08 (2012). Soil and rock samples will be discarded 60 days after the date of the final report unless otherwise directed.

SOIL CLASSIFICATION CHART

COARSE GRAINED SOILS	GRAVELS	Clean Gravel less than 5% fines	GW	Well graded gravel
			GP	Poorly graded gravel
		Gravels with Fines more than 12% fines	GM	Silty gravel
			GC	Clayey gravel
	SANDS	Clean Sand less than 5% fines	SW	Well graded sand
			SP	Poorly graded sand
		Sands with Fines more than 12% fines	SM	Silty sand
			SC	Clayey sand
FINE GRAINED SOILS	SILTS AND CLAYS Liquid Limit less than 50	Inorganic	CL	Lean clay
			ML	Silt
		Organic	OL	Organic clay and silt
			CH	Fat clay
	SILTS AND CLAYS Liquid Limit 50 or more	Inorganic	MH	Elastic silt
			OH	Organic clay and silt
HIGHLY ORGANIC SOILS		Organic matter, dark color, organic odor	PT	Peat

PARTICLE SIZE IDENTIFICATION

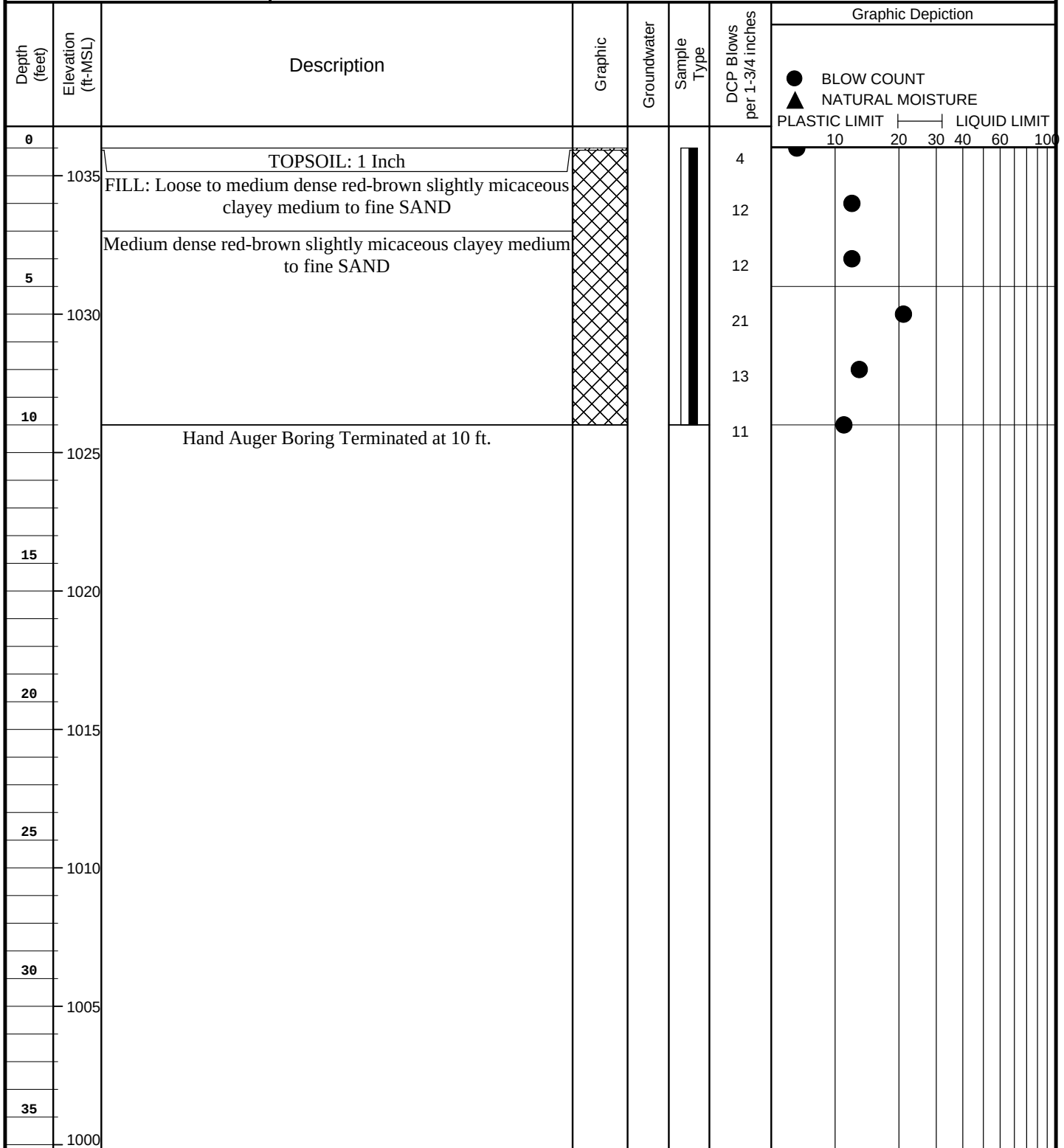
GRAVELS	Coarse	¾ inch to 3 inches
	Fine	No. 4 to ¾ inch
SANDS	Coarse	No. 10 to No. 4
	Medium	No. 40 to No. 10
	Fine	No. 200 to No. 40
SILTS AND CLAYS		Passing No. 200



HAND AUGER BORING RECORD B-1

PROJECT: Lidl-Norcross **PROJECT NO.:** 10103-2021006
CLIENT: Lidl US, LLC
PROJECT LOCATION: 4990 Jimmy Carter Blvd
LOCATION: Norcross, GA **ELEVATION:** 1036 Ft-MSL
DRILLER: NOVA **LOGGED BY:** MM
DRILLING METHOD: Hand Auger/DCP **DATE:** 3/3/2021
DEPTH TO - WATER> INITIAL: NE **AFTER 24 HOURS:** NE **CAVING>** C

This information pertains only to this boring and should not be interpreted as being indicative of the site.





HAND AUGER BORING RECORD B-2

PROJECT: Lidl-Norcross PROJECT NO.: 10103-2021006
CLIENT: Lidl US, LLC
PROJECT LOCATION: 4990 Jimmy Carter Blvd
LOCATION: Norcross, GA ELEVATION: 1036 Ft-MSL
DRILLER: NOVA LOGGED BY: MM
DRILLING METHOD: Hand Auger/DCP DATE: 3/2/2021
DEPTH TO - WATER> INITIAL: NE AFTER 24 HOURS: NE CAVING> C

This information pertains only to this boring and should not be interpreted as being indicative of the site.

Depth (feet)	Elevation (ft-MSL)	Description	Graphic	Groundwater	Sample Type	DCP Blows per 1-3/4 inches	Graphic Depiction									
							● BLOW COUNT ▲ NATURAL MOISTURE PLASTIC LIMIT LIQUID LIMIT 10 20 30 40 60 100									
0		TOPSOIL: 1 Inch														
	1035	FILL: Loose red-brown slightly micaceous clayey medium to fine SAND				7										
		Medium dense red-brown slightly micaceous clayey medium to fine SAND				17										
5		Hand Auger Refusal at 5 ft.														
	1030															
10																
	1025															
15																
	1020															
20																
	1015															
25																
	1010															
30																
	1005															
35																
	1000															



TEST BORING RECORD B-3

PROJECT: Lidl-Norcross

PROJECT NO.: 10103-2021006

CLIENT: Lidl US, LLC

PROJECT LOCATION: 4990 Jimmy Carter Blvd

LOCATION: Norcross, GA

ELEVATION: 1031 Ft-MSL

DRILLER: Premier Drilling, LLC

LOGGED BY: MM

DRILLING METHOD: Hollow Stem Auger

DATE: 3/2/2021

DEPTH TO - WATER> INITIAL: NE

AFTER 24 HOURS: NE

CAVING> C.

This information pertains only to this boring and should not be interpreted as being indicative of the site.

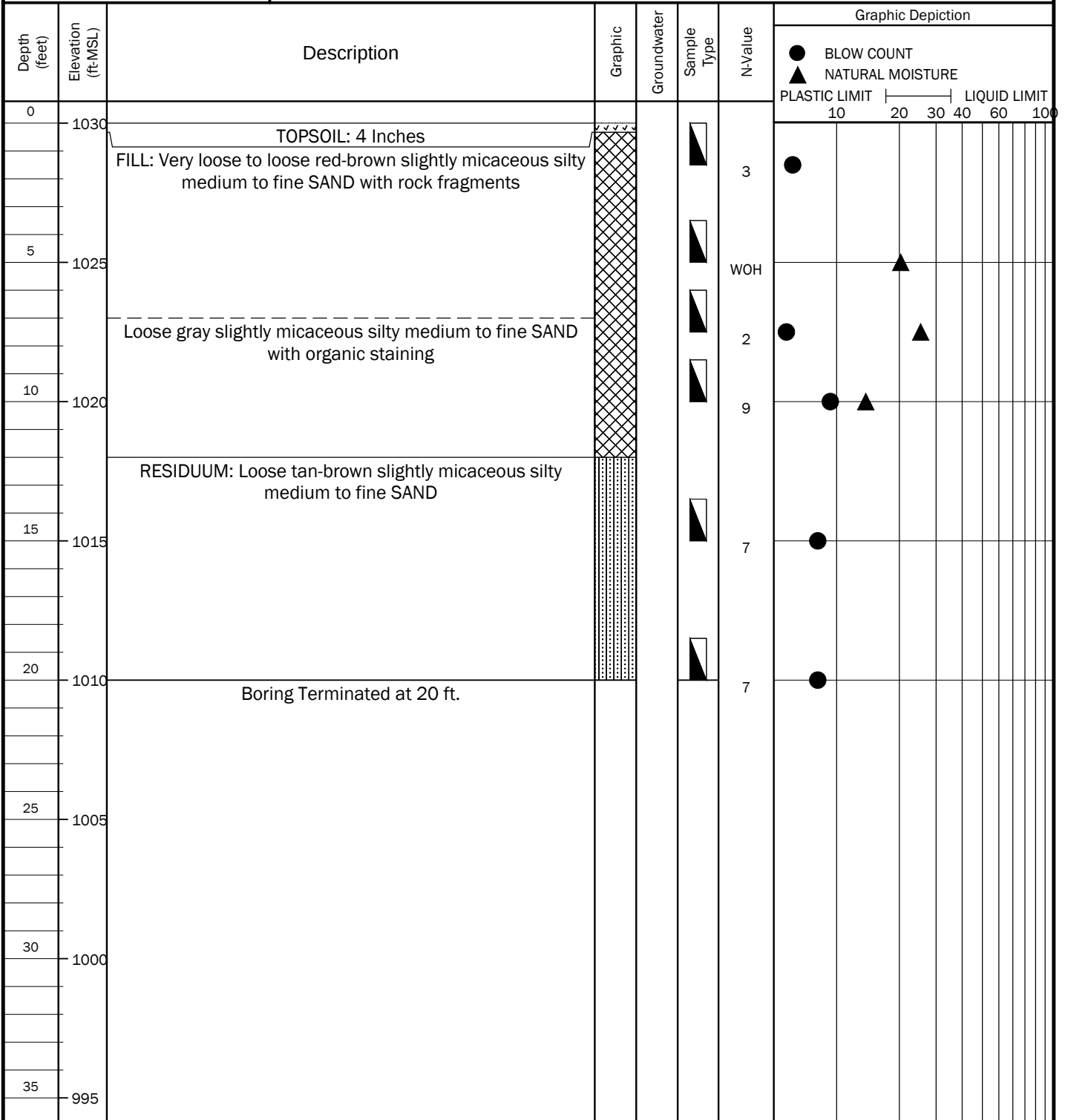
Depth (feet)	Elevation (ft-MSL)	Description	Graphic	Groundwater	Sample Type	N-Value	Graphic Depiction									



TEST BORING RECORD B-4

PROJECT: Lidl-NorcrossPROJECT NO.: 10103-2021006CLIENT: Lidl US, LLCPROJECT LOCATION: 4990 Jimmy Carter BlvdLOCATION: Norcross, GAELEVATION: 1030 Ft-MSLDRILLER: Premier Drilling, LLCLOGGED BY: MMDRILLING METHOD: Hollow Stem AugerDATE: 3/2/2021DEPTH TO - WATER> INITIAL: NENEAFTER 24 HOURS: NENECAVING> C

This information pertains only to this boring and should not be interpreted as being indicative of the site.

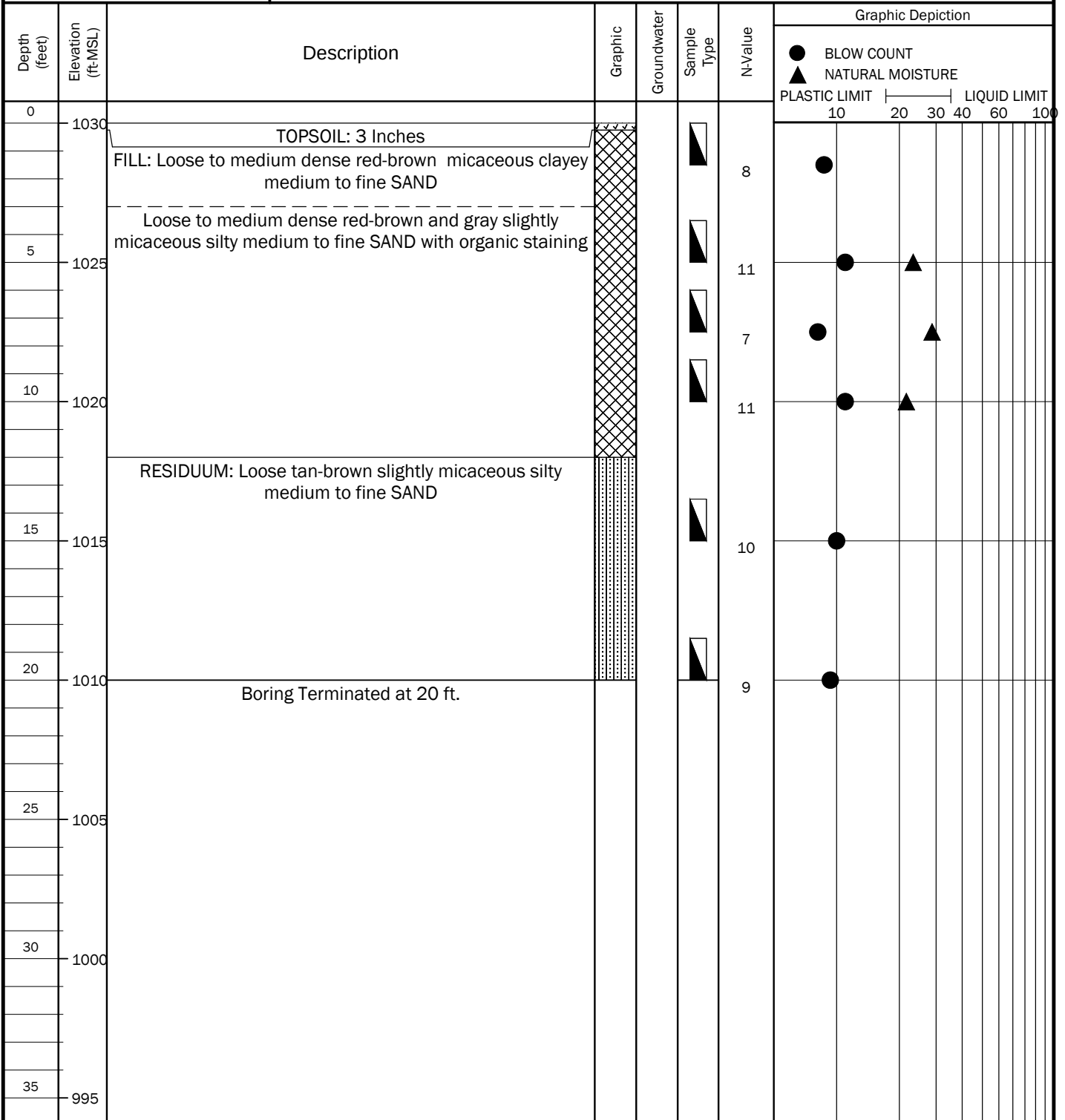




TEST BORING RECORD B-5

PROJECT: Lidl-NorcrossPROJECT NO.: 10103-2021006CLIENT: Lidl US, LLCPROJECT LOCATION: 4990 Jimmy Carter BlvdLOCATION: Norcross, GAELEVATION: 1030 Ft-MSLDRILLER: Premier Drilling, LLCLOGGED BY: MMDRILLING METHOD: Hollow Stem AugerDATE: 3/2/2021DEPTH TO - WATER> INITIAL: NEAFTER 24 HOURS: NECAVING> C

This information pertains only to this boring and should not be interpreted as being indicative of the site.



This information pertains only to this boring and should not be interpreted as being indicative of the site.



TEST BORING RECORD B-6

PROJECT: Lidl-Norcross PROJECT NO.: 10103-2021006
 CLIENT: Lidl US, LLC
 PROJECT LOCATION: 4990 Jimmy Carter Blvd
 LOCATION: Norcross, GA ELEVATION: 1026 Ft-MSL
 DRILLER: Premier Drilling, LLC LOGGED BY: MM
 DRILLING METHOD: Hollow Stem Auger DATE: 3/2/2021
 DEPTH TO - WATER> INITIAL: NE AFTER 24 HOURS: NE CAVING> C.

Depth (feet)	Elevation (ft-MSL)	Description	Graphic	Groundwater	Sample Type	N-Value	Graphic Depiction									
							● BLOW COUNT ▲ NATURAL MOISTURE PLASTIC LIMIT LIQUID LIMIT 10 20 30 40 60 100									
0																
		WOODCHIPS: 1 Foot														
	1025	FILL: Very loose to loose red-brown slightly micaceous clayey medium to fine SAND				2	●									
		Loose red-brown slightly micaceous silty medium to fine SAND														
5						3	●									
	1020															
						5	●									
10						5	●									
	1015															
		RESIDUUM: Loose red-brown slightly micaceous silty medium to fine SAND														
15						7	●									
	1010	Boring Terminated at 15 ft.														
20																
	1005															
25																
	1000															
30																
	995															
35																

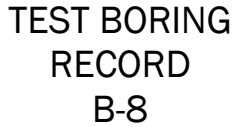


TEST BORING RECORD B-7

PROJECT: Lidl-Norcross PROJECT NO.: 10103-2021006
CLIENT: Lidl US, LLC
PROJECT LOCATION: 4990 Jimmy Carter Blvd
LOCATION: Norcross, GA ELEVATION: 1028 Ft-MSL
DRILLER: Premier Drilling, LLC LOGGED BY: MM
DRILLING METHOD: Hollow Stem Auger DATE: 3/2/2021
DEPTH TO - WATER> INITIAL: 7 AFTER 24 HOURS: 7 CAVING> C.

This information pertains only to this boring and should not be interpreted as being indicative of the site.

Depth (feet)	Elevation (ft-MSL)	Description	Graphic	Groundwater	Sample Type	N-Value	Graphic Depiction									
							● BLOW COUNT ▲ NATURAL MOISTURE PLASTIC LIMIT LIQUID LIMIT 10 20 30 40 60 100									
0		WOODCHIPS: 1 Foot														
		FILL: Very loose red-brown slightly micaceous clayey medium to fine SAND				4	●									
1025		Loose red-brown slightly micaceous silty medium to fine SAND				4	●									
5						5	●									
1020						6	●									
10						5	●									
1015																
15		Boring Terminated at 15 ft.				5	●									
1010																
20																
1005																
25																
1000																
30																
995																
35																

PROJECT NO.: 10103-2021006

PROJECT LOCATION: 4990 Jimmy Carter Blvd

ELEVATION: 1029 Ft-MSL

LOGGED BY: _____ MM

DATE: 3/2/2021

DEPTH TO - WATER> INITIAL: NE AFTER 24 HOURS: NE CAVING> C

Depth (feet)	Elevation (ft-MSL)	Description	Graphic	Groundwater	Sample Type	N-Value	Graphic Depiction	
							● BLOW COUNT ▲ NATURAL MOISTURE	PLASTIC LIMIT LIQUID LIMIT
0		TOPSOIL: 2 Inches						
		FILL: Loose red-brown slightly micaceous clayey medium to fine SAND with rock fragments						
	1025	Very loose to loose red-brown slightly micaceous silty medium to fine SAND						
5						5	●	
						2	●	▲
						8	●	▲
10	1020					9	●	▲
	1015	RESIDUUM: Loose tan-brown slightly micaceous silty medium to fine SAND				8	●	
15								
	1010					7	●	
20		Boring Terminated at 20 ft.						
25	1005							
30	1000							
35	995							

APPENDIX C

Laboratory Test Results

COMPACTION TEST REPORT

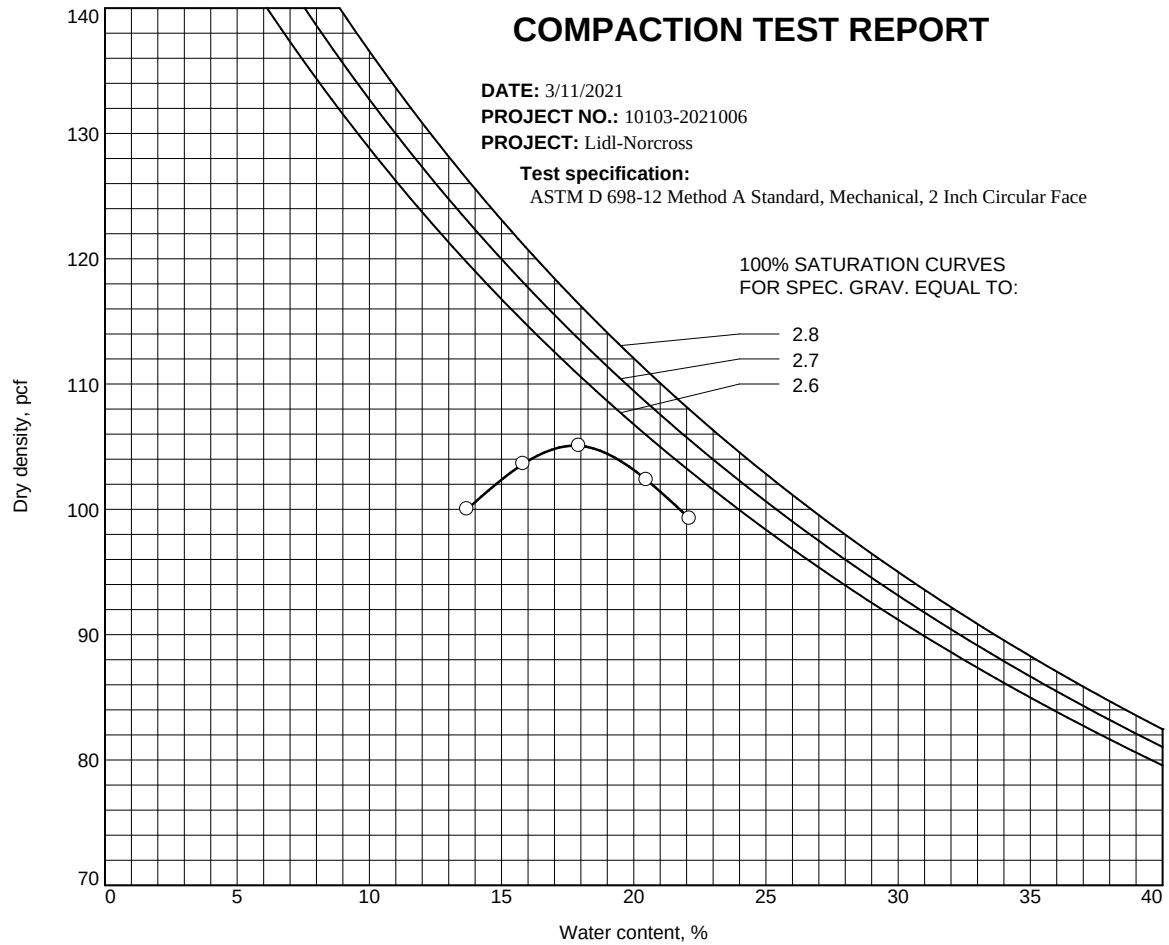
DATE: 3/11/2021

PROJECT NO.: 10103-2021006

PROJECT: Lidl-Norcross

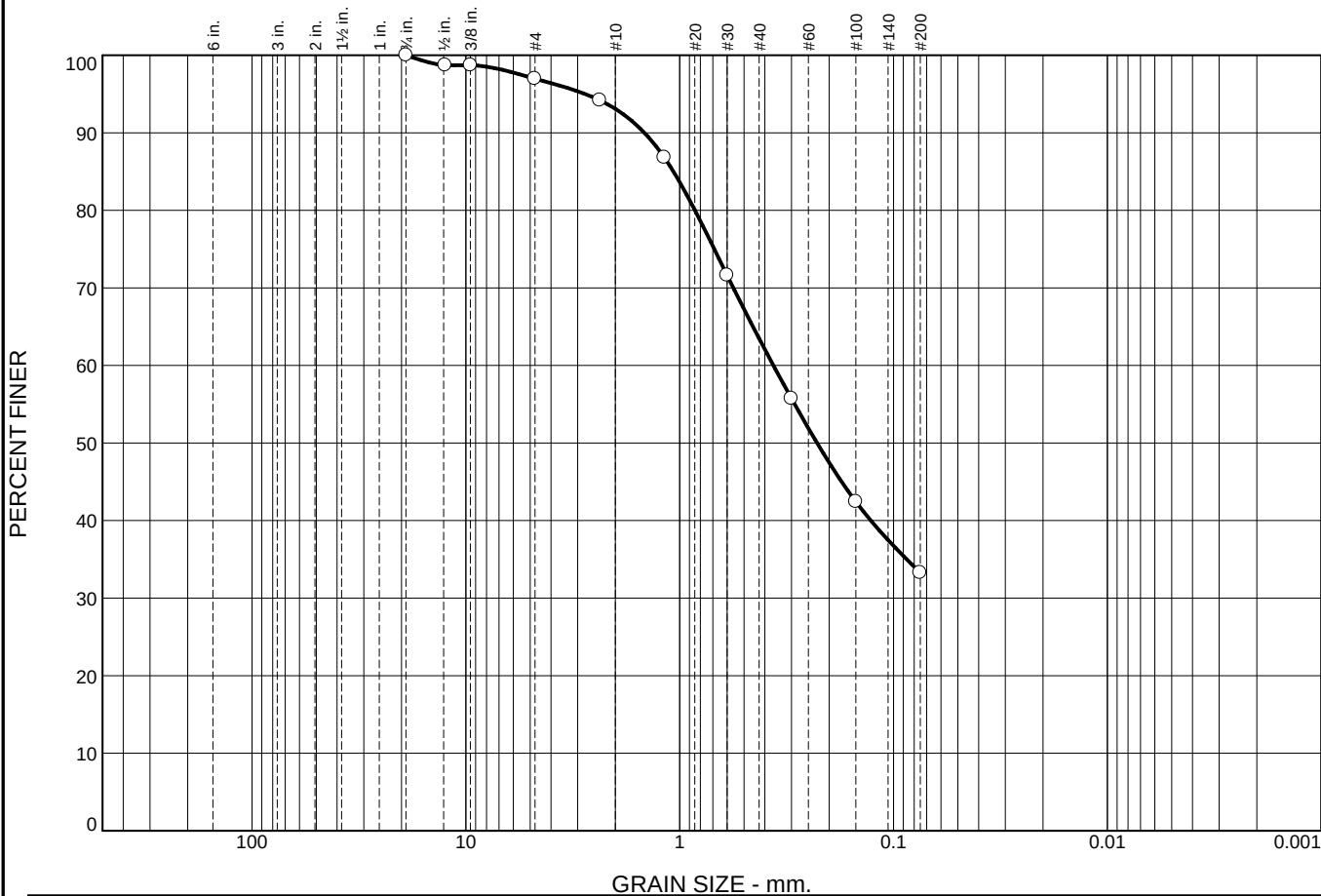
Test specification:

ASTM D 698-12 Method A Standard, Mechanical, 2 Inch Circular Face



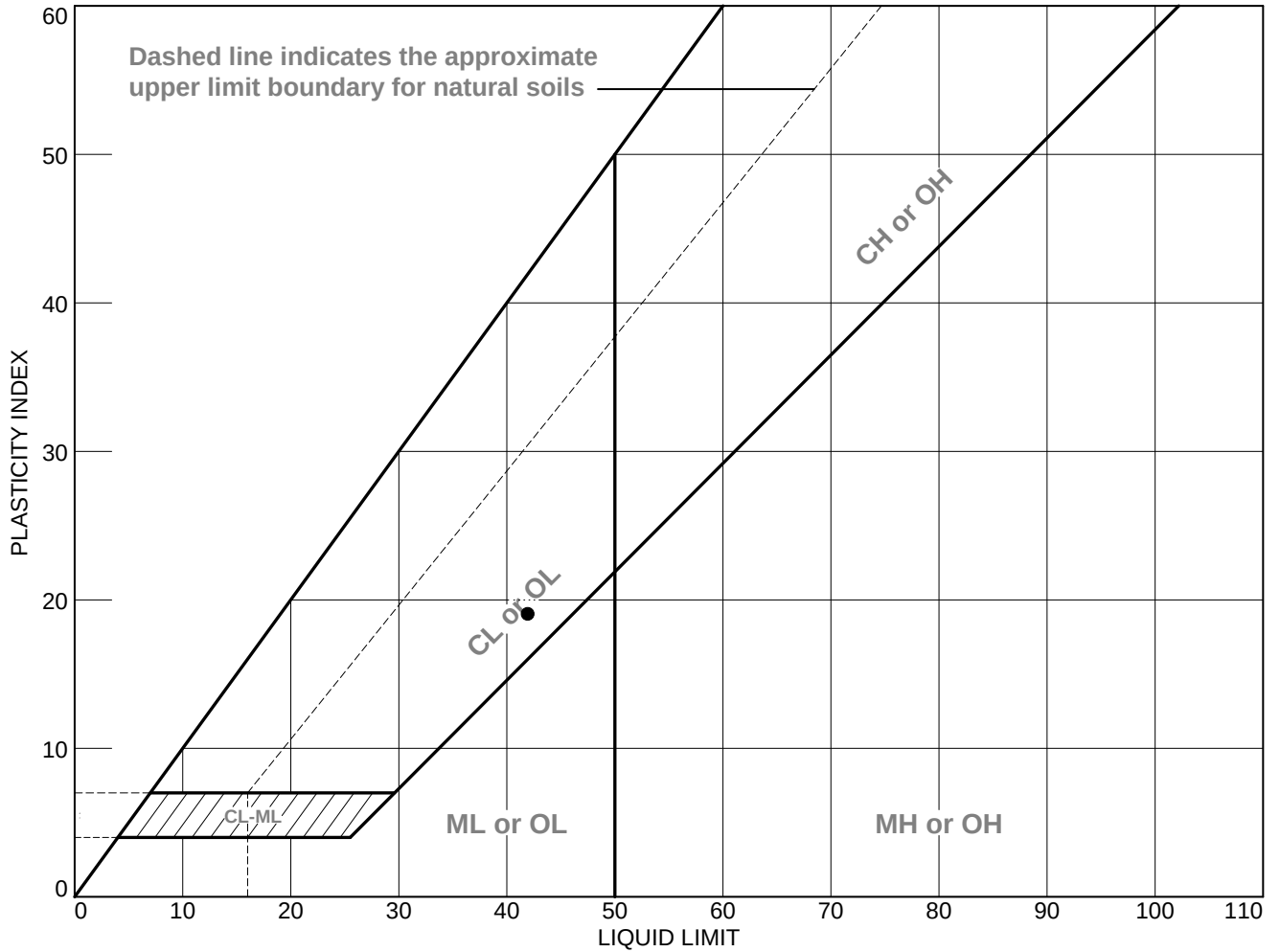
	Sample No.	Elev. or Depth	Material Description	Specific Gravity	LL	PL	Oversize	% < #200
○	P-01	0.5-1.5 Ft	Red-brown slightly micaceous clayey medium to fine SAND		42	23	%>#4=3.0	33.3
Sample No.			P-01					
Natural water content, percent			18.6					
Optimum water content, percent			17.8					
Max dry density, pcf			105.1					
Remarks:			Project: Lidl-Norcross			Project No.: 10103-2021006		
			Location:					
			Source: B-1					
			Nova Engineering & Environmental Norcross, GA					
Figure								

Particle Size Distribution Report



% +3"		% Gravel		% Sand			% Fines			
		Coarse	Fine	Coarse	Medium	Fine	Silt	Clay		
☐	0.0	0.0	3.0	3.9	29.6	30.2	33.3			
☐										
☒	LL	PL	D ₈₅	D ₆₀	D ₅₀	D ₃₀	D ₁₅	D ₁₀	C _c	C _u
☐	42	23	1.0694	0.3645	0.2278					
☐										
MATERIAL DESCRIPTION							TEST DATE	USCS	NM	
☐ Red-brown slightly micaceous clayey medium to fine SAND								SC	18.6	
Project No. 10103- Client: Lidl US, LLC							Remarks:			
Project: Lidl-Norcross										
☐ Source of Sample: B-1 Depth: 0.5-1.5 Ft Sample Number: P-01										
Nova Engineering & Environmental							Figure			
Norcross, GA										

LIQUID AND PLASTIC LIMITS TEST REPORT



SOIL DATA								
SYMBOL	SOURCE	SAMPLE NO.	DEPTH	NATURAL WATER CONTENT (%)	PLASTIC LIMIT (%)	LIQUID LIMIT (%)	PLASTICITY INDEX (%)	USCS
●	B-1	P-01	0.5-1.5 Ft	18.6	23	42	19	SC

**Nova Engineering
& Environmental
Norcross, GA**

Client: Lidl US, LLC
Project: Lidl-Norcross

Project No.: 10103-2021006

Figure

APPENDIX D

Qualifications of Recommendations

QUALIFICATIONS OF RECOMMENDATIONS

The findings, conclusions and recommendations presented in this report represent our professional opinions concerning subsurface conditions at the site. The opinions presented are relative to the dates of our site work and should not be relied on to represent conditions at later dates or at locations not explored. The opinions included herein are based on information provided to us, the data obtained at specific locations during the study and our experience. If additional information becomes available that might impact our geotechnical opinions, it will be necessary for NOVA to review the information, reassess the potential concerns, and re-evaluate our conclusions and recommendations.

Regardless of the thoroughness of a geotechnical exploration, there is the possibility that conditions between borings will differ from those encountered at specific boring locations, that conditions are not as anticipated by the designers and/or the contractors, or that either natural events or the construction process have altered the subsurface conditions. These variations are an inherent risk associated with subsurface conditions in this region and the approximate methods used to obtain the data. These variations may not be apparent until construction.

The professional opinions presented in this geotechnical report are not final. Field observations and foundation installation monitoring by the geotechnical engineer, as well as soil density testing and other quality assurance functions associated with site earthwork and foundation construction, are an extension of this report. Therefore, NOVA should be retained by the owner to observe all earthwork and foundation construction to document that the conditions anticipated in this study actually exist, and to finalize or amend our conclusions and recommendations. NOVA is not responsible or liable for the conclusions and recommendations presented in this report if NOVA does not perform these observation and testing services.

This report is intended for the sole use of the CLIENT only. The scope of work performed during this study was developed for purposes specifically intended by the CLIENT and may not satisfy other users' requirements. Use of this report or the findings, conclusions or recommendations by others will be at the sole risk of the user. NOVA is not responsible or liable for the interpretation by others of the data in this report, nor their conclusions, recommendations or opinions.

Our professional services have been performed, our findings obtained, our conclusions derived, and our recommendations prepared in accordance with generally accepted geotechnical engineering principles and practices in the State of Georgia. This warranty is in lieu of all other statements or warranties, either expressed or implied.

Important Information about This Geotechnical-Engineering Report

Subsurface problems are a principal cause of construction delays, cost overruns, claims, and disputes.

While you cannot eliminate all such risks, you can manage them. The following information is provided to help.

Geotechnical Services Are Performed for Specific Purposes, Persons, and Projects

Geotechnical engineers structure their services to meet the specific needs of their clients. A geotechnical-engineering study conducted for a civil engineer may not fulfill the needs of a constructor — a construction contractor — or even another civil engineer. Because each geotechnical-engineering study is unique, each geotechnical-engineering report is unique, prepared *solely* for the client. No one except you should rely on this geotechnical-engineering report without first conferring with the geotechnical engineer who prepared it. *And no one — not even you — should apply this report for any purpose or project except the one originally contemplated.*

Read the Full Report

Serious problems have occurred because those relying on a geotechnical-engineering report did not read it all. Do not rely on an executive summary. Do not read selected elements only.

Geotechnical Engineers Base Each Report on a Unique Set of Project-Specific Factors

Geotechnical engineers consider many unique, project-specific factors when establishing the scope of a study. Typical factors include: the client's goals, objectives, and risk-management preferences; the general nature of the structure involved, its size, and configuration; the location of the structure on the site; and other planned or existing site improvements, such as access roads, parking lots, and underground utilities. Unless the geotechnical engineer who conducted the study specifically indicates otherwise, do not rely on a geotechnical-engineering report that was:

- not prepared for you;
- not prepared for your project;
- not prepared for the specific site explored; or
- completed before important project changes were made.

Typical changes that can erode the reliability of an existing geotechnical-engineering report include those that affect:

- the function of the proposed structure, as when it's changed from a parking garage to an office building, or from a light-industrial plant to a refrigerated warehouse;
- the elevation, configuration, location, orientation, or weight of the proposed structure;
- the composition of the design team; or
- project ownership.

As a general rule, *always* inform your geotechnical engineer of project changes—even minor ones—and request an

assessment of their impact. *Geotechnical engineers cannot accept responsibility or liability for problems that occur because their reports do not consider developments of which they were not informed.*

Subsurface Conditions Can Change

A geotechnical-engineering report is based on conditions that existed at the time the geotechnical engineer performed the study. *Do not rely on a geotechnical-engineering report whose adequacy may have been affected by:* the passage of time; man-made events, such as construction on or adjacent to the site; or natural events, such as floods, droughts, earthquakes, or groundwater fluctuations. *Contact the geotechnical engineer before applying this report to determine if it is still reliable.* A minor amount of additional testing or analysis could prevent major problems.

Most Geotechnical Findings Are Professional Opinions

Site exploration identifies subsurface conditions only at those points where subsurface tests are conducted or samples are taken. Geotechnical engineers review field and laboratory data and then apply their professional judgment to render an opinion about subsurface conditions throughout the site. Actual subsurface conditions may differ — sometimes significantly — from those indicated in your report. Retaining the geotechnical engineer who developed your report to provide geotechnical-construction observation is the most effective method of managing the risks associated with unanticipated conditions.

A Report's Recommendations Are Not Final

Do not overrely on the confirmation-dependent recommendations included in your report. *Confirmation-dependent recommendations are not final*, because geotechnical engineers develop them principally from judgment and opinion. Geotechnical engineers can finalize their recommendations *only* by observing actual subsurface conditions revealed during construction. *The geotechnical engineer who developed your report cannot assume responsibility or liability for the report's confirmation-dependent recommendations if that engineer does not perform the geotechnical-construction observation required to confirm the recommendations' applicability.*

A Geotechnical-Engineering Report Is Subject to Misinterpretation

Other design-team members' misinterpretation of geotechnical-engineering reports has resulted in costly

problems. Confront that risk by having your geotechnical engineer confer with appropriate members of the design team after submitting the report. Also retain your geotechnical engineer to review pertinent elements of the design team's plans and specifications. Constructors can also misinterpret a geotechnical-engineering report. Confront that risk by having your geotechnical engineer participate in prebid and preconstruction conferences, and by providing geotechnical construction observation.

Do Not Redraw the Engineer's Logs

Geotechnical engineers prepare final boring and testing logs based upon their interpretation of field logs and laboratory data. To prevent errors or omissions, the logs included in a geotechnical-engineering report should *never* be redrawn for inclusion in architectural or other design drawings. Only photographic or electronic reproduction is acceptable, *but recognize that separating logs from the report can elevate risk.*

Give Constructors a Complete Report and Guidance

Some owners and design professionals mistakenly believe they can make constructors liable for unanticipated subsurface conditions by limiting what they provide for bid preparation. To help prevent costly problems, give constructors the complete geotechnical-engineering report, *but* preface it with a clearly written letter of transmittal. In that letter, advise constructors that the report was not prepared for purposes of bid development and that the report's accuracy is limited; encourage them to confer with the geotechnical engineer who prepared the report (a modest fee may be required) and/or to conduct additional study to obtain the specific types of information they need or prefer. A prebid conference can also be valuable. *Be sure constructors have sufficient time to perform additional study.* Only then might you be in a position to give constructors the best information available to you, while requiring them to at least share some of the financial responsibilities stemming from unanticipated conditions.

Read Responsibility Provisions Closely

Some clients, design professionals, and constructors fail to recognize that geotechnical engineering is far less exact than other engineering disciplines. This lack of understanding has created unrealistic expectations that have led to disappointments, claims, and disputes. To help reduce the risk of such outcomes, geotechnical engineers commonly include a variety of explanatory provisions in their reports. Sometimes labeled "limitations," many of these provisions indicate where geotechnical engineers' responsibilities begin and end, to help

others recognize their own responsibilities and risks. *Read these provisions closely.* Ask questions. Your geotechnical engineer should respond fully and frankly.

Environmental Concerns Are Not Covered

The equipment, techniques, and personnel used to perform an *environmental* study differ significantly from those used to perform a *geotechnical* study. For that reason, a geotechnical-engineering report does not usually relate any environmental findings, conclusions, or recommendations; e.g., about the likelihood of encountering underground storage tanks or regulated contaminants. *Unanticipated environmental problems have led to numerous project failures.* If you have not yet obtained your own environmental information, ask your geotechnical consultant for risk-management guidance. *Do not rely on an environmental report prepared for someone else.*

Obtain Professional Assistance To Deal with Mold

Diverse strategies can be applied during building design, construction, operation, and maintenance to prevent significant amounts of mold from growing on indoor surfaces. To be effective, all such strategies should be devised for the *express purpose* of mold prevention, integrated into a comprehensive plan, and executed with diligent oversight by a professional mold-prevention consultant. Because just a small amount of water or moisture can lead to the development of severe mold infestations, many mold-prevention strategies focus on keeping building surfaces dry. While groundwater, water infiltration, and similar issues may have been addressed as part of the geotechnical-engineering study whose findings are conveyed in this report, the geotechnical engineer in charge of this project is not a mold prevention consultant; *none of the services performed in connection with the geotechnical engineer's study were designed or conducted for the purpose of mold prevention. Proper implementation of the recommendations conveyed in this report will not of itself be sufficient to prevent mold from growing in or on the structure involved.*

Rely, on Your GBC-Member Geotechnical Engineer for Additional Assistance

Membership in the Geotechnical Business Council of the Geoprofessional Business Association exposes geotechnical engineers to a wide array of risk-confrontation techniques that can be of genuine benefit for everyone involved with a construction project. Confer with you GBC-Member geotechnical engineer for more information.



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